# 🎭 FunnyAuth — Login & Registration Page

> একটি মজাদার, অ্যানিমেটেড Login/Register পেজ — Next.js + TypeScript + MongoDB

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## 📌 Project Overview

এই প্রজেক্টটি হলো একটি **Single-Page Authentication System** যেখানে Login এবং Register দুটো ফর্মই একই পেজে থাকবে। ফর্ম সুইচ করার সময় **smooth slide animation**, **color theme change**, এবং **funny side messages** দেখাবে। সব কিছু হবে **Next.js (App Router) + TypeScript + MongoDB** দিয়ে।

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## 🧱 Tech Stack (সম্পূর্ণ তালিকা)

| Layer | Technology | কারণ |
|---|---|---|
| **Framework** | Next.js 14+ (App Router) | SSR, API Routes, file-based routing |
| **Language** | TypeScript | Type safety, ভুল কমানো |
| **Styling** | Tailwind CSS + CSS Modules | Utility-first + custom animations |
| **Animation** | Framer Motion | Smooth slide, fade, spring animations |
| **Form Handling** | React Hook Form | Performance, validation |
| **Validation** | Zod | Schema-based form validation |
| **Database** | MongoDB Atlas | Cloud MongoDB |
| **ODM** | Mongoose | MongoDB object modeling |
| **Auth** | NextAuth.js v5 (Auth.js) | Session management, JWT |
| **Password** | bcryptjs | Password hashing |
| **Toast/Alert** | react-hot-toast | Funny notification messages |
| **Icons** | Lucide React | Clean icon library |
| **Fonts** | Google Fonts (Bangers + Nunito) | Playful + readable |
| **Environment** | dotenv (.env.local) | Secret management |
| **HTTP Client** | Axios | API calls from frontend |

| Layer | Technology | কারণ |
|---|---|---|
| **Dev Tools** | ESLint, Prettier, Husky | Code quality |

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## 📁 Folder Structure

```
funny-auth/
├── app/
│   ├── api/
│   │   └── auth/
│   │       ├── register/
│   │       │   └── route.ts          ← Registration API endpoint
│   │       └── [...nextauth]/
│   │           └── route.ts          ← NextAuth handler
│   ├── (auth)/
│   │   └── page.tsx                  ← Main Login/Register page
│   ├── dashboard/
│   │   └── page.tsx                  ← After login redirect
│   ├── layout.tsx
│   └── globals.css
│
├── components/
│   ├── auth/
│   │   ├── AuthWrapper.tsx           ← মূল container (দুই ভাগ)
│   │   ├── LoginForm.tsx             ← Login form component
│   │   ├── RegisterForm.tsx          ← Register form component
│   │   ├── SidePanel.tsx             ← বিপরীত পাশের funny message panel
│   │   └── FormToggleButton.tsx      ← Toggle link/button
│   ├── ui/
│   │   ├── InputField.tsx            ← Reusable animated input
│   │   ├── SubmitButton.tsx          ← Loading state button
│   │   └── PasswordStrength.tsx      ← Register এ password strength bar
│   └── animations/
│       ├── SlideTransition.tsx       ← Framer Motion slide wrapper
│       └── FloatingEmojis.tsx        ← Background funny emoji particles
│
├── lib/
│   ├── mongodb.ts                    ← MongoDB connection singleton
│   ├── auth.ts                       ← NextAuth config
│   └── validations/
│       ├── loginSchema.ts            ← Zod login schema
```

```
│       └── registerSchema.ts          ← Zod register schema
│
├── models/
│   └── User.ts                        ← Mongoose User model
│
├── hooks/
│   ├── useAuthToggle.ts               ← Login/Register toggle state
│   └── useAnimatedForm.ts             ← Form animation states
│
├── types/
│   └── auth.types.ts                  ← TypeScript interfaces
│
├── constants/
│   └── funnyMessages.ts               ← Funny messages list
│
├── .env.local                         ← Environment variables
├── tailwind.config.ts
├── next.config.ts
└── tsconfig.json
```

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## 🎨 UI/UX Design Specification

### Layout (Desktop)

```
┌──────────────────────────────────────────────────────────────┐
│                                                              │
│   ┌──────────────────┐    ┌──────────────────────────┐      │
│   │                  │    │                          │      │
│   │   LOGIN FORM     │    │   SIDE PANEL             │      │
│   │   (বাম পাশ)      │    │   (ডান পাশ)              │      │
│   │                  │    │   Funny Messages +       │      │
│   │   📧 Email       │    │   Illustrations          │      │
│   │   🔒 Password    │    │                          │      │
│   │   [Login Button] │    │   "Welcome back!         │      │
│   │                  │    │   We missed you 🥳"      │      │
│   │   → Register Link │    │                          │      │
│   │                  │    │                          │      │
│   └──────────────────┘    └──────────────────────────┘      │
│                                                              │
└──────────────────────────────────────────────────────────────┘
```

```
⬇️ Register লিংক ক্লিক করলে ⬇️

┌──────────────────────────────────────────────────────────┐
│                                                          │
│   ┌──────────────────────┐   ┌──────────────────────┐    │
│   │                      │   │                      │    │
│   │  SIDE PANEL          │   │  REGISTER FORM       │    │
│   │  (বাম পাশে চলে আসে)  │   │  (ডান পাশে)          │    │
│   │                      │   │                      │    │
│   │ "Join the chaos!     │   │  👤 Name             │    │
│   │  We promise we're    │   │  📧 Email            │    │
│   │  mostly harmless 😈" │   │  🔒 Password         │    │
│   │                      │   │  🔒 Confirm          │    │
│   │                      │   │  [Register]          │    │
│   │                      │   │                      │    │
│   │                      │   │  ← Login Link        │    │
│   │                      │   │                      │    │
│   └──────────────────────┘   └──────────────────────┘    │
│                                                          │
└──────────────────────────────────────────────────────────┘
```

### Color Themes

| State | Primary Color | Accent | Background |
|---|---|---|---|
| **Login Mode** | `#6C63FF` (Purple) | `#FF6584` (Pink) | Deep dark `#0D0D1A` |
| **Register Mode** | `#FF6B35` (Orange) | `#FFE66D` (Yellow) | Warm dark `#1A0D00` |

> Theme transition হবে `CSS custom properties` + Framer Motion `animate` দিয়ে — smooth 0.6s ease।

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## ✨ Animation Specification

### 1. Form Slide Animation (সবচেয়ে গুরুত্বপূর্ণ)

```
Login → Register:
  - Form: slide out to LEFT (x: 0 → -100%) + fade out
  - Register Form: slide in from RIGHT (x: +100% → 0) + fade in
  - Side Panel: slide from RIGHT to LEFT simultaneously
  - Duration: 0.5s spring animation
```

### 2. Input Field Micro-interactions

- Focus হলে: label ওপরে উঠবে + border glow
- Error হলে: shake animation (left-right wobble)
- Success হলে: green checkmark pop-in

### 3. Submit Button States

```
Idle      → "Login 🚀"
Loading   → Spinning emoji "⏳ Logging in..."
Success   → "✅ Welcome back!" + confetti burst
Error     → "😅 Oops! Try again"
```

### 4. Background Floating Emojis

Register পেজে background এ floating emojis: 🎉 🎊 🥳 ✨ 🚀
Login পেজে: 👋 🔐 💻 🌙 ⭐

### 5. Side Panel Message Typewriter Effect

প্রতিটি funny message typewriter animation দিয়ে লেখা হবে।

### 6. Password Strength Bar (Register only)

```
Weak     → Red bar 25%    + message: "Seriously? '123456'? 😒"
Fair     → Yellow bar 50% + message: "Getting warmer... 🤔"
Good     → Blue bar 75%   + message: "Nice! Almost there 💪"
Strong   → Green bar 100% + message: "Fort Knox approved! 🏰"
```

---

## 😂 Funny Messages List

### Login Side Panel Messages (ডান পাশে)

```typescript
```

```
const loginMessages = [
  {
    title: "Welcome Back, Legend! 👑",
    subtitle: "We kept your seat warm. Well... it's the internet. But still."
  },
  {
    title: "Oh, You're Back? 😌",
    subtitle: "We were JUST talking about you. (We weren't. But hello!)"
  },
  {
    title: "The Prodigal User Returns 🍻",
    subtitle: "Your account missed you. It cried a little. Don't tell it we told you
  }
];
```

### Register Side Panel Messages (বাম পাশে)

typescript

```
const registerMessages = [
  {
    title: "Join The Club! 🎪",
    subtitle: "We have cookies. Well, browser cookies. But close enough!"
  },
  {
    title: "Creating Your Digital Alter Ego 🦸",
    subtitle: "Warning: Side effects may include productivity and mild fun."
  },
  {
    title: "Almost Human... er, Registered! 🤖",
    subtitle: "Fill in the form. We promise to spam you responsibly."
  }
];
```

### Toast Notifications

typescript

```
// Successful login
toast.success("You're in! Try not to break anything 😅")

// Failed login
toast.error("Nope! Password wrong. Did you try 'password123'? 😂")

// Successful register
toast.success("Welcome to the chaos! 🎉 Check your email... or don't.")

// Email already exists
toast.error("Someone already stole that email. Classic. 😤")

// Network error
toast.error("Our servers are crying 😭 Try again?")
```

---

## 🗄️ Database Schema (MongoDB)

### User Model (`models/User.ts`)

```typescript
interface IUser {
  _id: ObjectId;
  name: string;                // Full name
  email: string;               // Unique, lowercase
  password: string;            // bcrypt hashed (saltRounds: 12)
  avatar?: string;             // Optional profile pic URL
  isVerified: boolean;         // Email verification (future)
  loginCount: number;          // কতবার login করেছে
  lastLogin?: Date;            // Last login timestamp
  funnyTitle?: string;         // "The Legendary User" etc.
  createdAt: Date;
  updatedAt: Date;
}
```

### MongoDB Indexes

```typescript
```

```
// Unique index on email
UserSchema.index({ email: 1 }, { unique: true });

// TTL index for unverified accounts (optional)
UserSchema.index({ createdAt: 1 }, { expireAfterSeconds: 86400 });
```

---

## 🔐 Authentication Flow

### Registration Flow

```
User fills form
    ↓
Zod validation (client-side)
    ↓
POST /api/auth/register
    ↓
Server-side Zod validation
    ↓
Check if email exists in MongoDB
    ↓
Hash password with bcrypt (saltRounds: 12)
    ↓
Save User to MongoDB
    ↓
Auto-login via NextAuth signIn()
    ↓
Redirect to /dashboard
    ↓
Show funny success toast 🎉
```

### Login Flow

```
User fills form
    ↓
Zod validation (client-side)
    ↓
NextAuth signIn('credentials', {...})
    ↓
NextAuth calls authorize() callback
    ↓
```

```
Find user in MongoDB by email
    ↓
bcrypt.compare(password, hash)
    ↓
Return user object → JWT created
    ↓
Redirect to /dashboard
    ↓
Show funny welcome toast 👑
```

---

## 🌐 API Routes

`POST /api/auth/register`

**Request Body:**

```json
{
  "name": "John Doe",
  "email": "john@example.com",
  "password": "SuperSecret123!",
  "confirmPassword": "SuperSecret123!"
}
```

**Success Response (201):**

```json
{
  "success": true,
  "message": "Account created! Welcome to the madness 🎉",
  "user": {
    "id": "64abc...",
    "name": "John Doe",
    "email": "john@example.com"
  }
}
```

**Error Responses:**

```json
```

```
// 400 - Validation Error
{ "success": false, "error": "Password must be at least 8 characters" }

// 409 - Email Exists
{ "success": false, "error": "Email already taken! Sneaky." }

// 500 - Server Error
{ "success": false, "error": "Server went on vacation 🏖️" }
```

---

## 📦 Installation Guide (Developer এর জন্য)

### Step 1: Project Setup

```bash
# Next.js প্রজেক্ট তৈরি
npx create-next-app@latest funny-auth --typescript --tailwind --app --src-dir

cd funny-auth
```

### Step 2: Dependencies Install

```bash
# Core dependencies
npm install mongoose bcryptjs next-auth@beta

# Form & Validation
npm install react-hook-form zod @hookform/resolvers

# Animation
npm install framer-motion

# UI utilities
npm install react-hot-toast lucide-react axios

# Type definitions
npm install -D @types/bcryptjs
```

### Step 3: Environment Variables (`.env.local`)

```env
# MongoDB
MONGODB_URI=mongodb+srv://<username>:<password>@cluster0.xxxxx.mongodb.net/funny-aut

# NextAuth
NEXTAUTH_URL=http://localhost:3000
NEXTAUTH_SECRET=your-super-secret-key-here-min-32-chars

# App
NEXT_PUBLIC_APP_NAME=FunnyAuth
NEXT_PUBLIC_APP_URL=http://localhost:3000
```

### Step 4: Run Development Server

```bash
npm run dev
# http://localhost:3000 এ open করো
```

---

## 🎯 Component Implementation Guide

### `AuthWrapper.tsx` — মূল কাঠামো

```tsx
// দুটো অংশ: formSide + panelSide
// isLogin state দিয়ে কোনটা কোথায় থাকবে নিয়ন্ত্রণ
// Framer Motion AnimatePresence দিয়ে smooth switching
// Background gradient theme change on toggle
```

### `LoginForm.tsx` — Login Form

```
Fields:
  - Email (type="email", autocomplete="email")
  - Password (show/hide toggle 👁️)

Buttons:
  - Submit: "Login 🚀" → loading → success/error
```

```
Links:
  - "Don't have an account? Join the fun →"
  - "Forgot password? (Don't worry, we forget things too 😅)"

Validation:
  - Email: valid format
  - Password: minimum 8 characters
```

### `RegisterForm.tsx` — Register Form

```
Fields:
  - Full Name
  - Email
  - Password (with strength indicator)
  - Confirm Password

Buttons:
  - Submit: "Create Account 🎉"

Links:
  - "Already one of us? Login →"

Extra:
  - Password strength bar (4 levels)
  - Real-time validation messages (funny)
```

### `SidePanel.tsx` — Funny Message Panel

```
- Background: gradient mesh
- Floating emoji particles
- Typewriter animated title
- Subtitle fade-in
- Decorative SVG illustration
- Changes content based on isLogin state
```

---

## 📱 Responsive Design

| Breakpoint | Layout |
|---|---|
| **Mobile** (< 768px) | Stack vertically, panel ওপরে form নিচে |
| **Tablet** (768px - 1024px) | Side by side, 40/60 split |
| **Desktop** (> 1024px) | Side by side, 50/50 split |

Mobile তে slide animation হবে **up/down** বরং **left/right** এর বদলে।

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## ♿ Accessibility

- সব form fields এ `aria-label` এবং `aria-describedby`
- Error messages `role="alert"` দিয়ে
- Keyboard navigation সম্পূর্ণ কাজ করবে
- Focus ring visible থাকবে
- Color contrast WCAG AA compliant
- `prefers-reduced-motion` respect করবে

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## 🔒 Security Checklist

- ☑ Password bcrypt hashing (saltRounds: 12)
- ☑ JWT secret minimum 32 characters
- ☑ Input sanitization (Zod validation)
- ☑ Rate limiting on auth endpoints (next-rate-limit)
- ☑ HTTPS only in production
- ☑ HTTP-only cookies for session
- ☑ CSRF protection (NextAuth built-in)
- ☑ MongoDB injection prevention (Mongoose)
- ☑ Environment variables never exposed to client
- ☑ Error messages না দেওয়া sensitive info

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## 🚀 Deployment

### Vercel (Recommended)

```bash
# Vercel CLI install
npm i -g vercel

# Deploy
vercel

# Environment variables Vercel dashboard এ add করো:
# MONGODB_URI, NEXTAUTH_SECRET, NEXTAUTH_URL
```

### MongoDB Atlas Setup

1. mongodb.com/atlas এ account তৈরি
2. Free tier cluster তৈরি (M0)
3. Database user তৈরি
4. Network access: `0.0.0.0/0` (production এ restrict করো)
5. Connection string copy করে `.env.local` এ paste করো

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## 📋 Developer Task Breakdown

### Phase 1: Foundation (1-2 দিন)

- [ ] Next.js project setup with TypeScript
- [ ] MongoDB connection (`lib/mongodb.ts`)
- [ ] User model তৈরি (`models/User.ts`)
- [ ] NextAuth configuration
- [ ] Zod validation schemas

### Phase 2: API Routes (1 দিন)

- [ ] Registration API route
- [ ] NextAuth login handler
- [ ] Error handling middleware

### Phase 3: UI Components (2-3 দিন)

- [ ] `AuthWrapper` with state management

- [ ] `LoginForm` with validation
- [ ] `RegisterForm` with password strength
- [ ] `SidePanel` with funny messages
- [ ] Reusable `InputField`, `SubmitButton`

### Phase 4: Animations (1-2 দিন)

- [ ] Framer Motion slide transitions
- [ ] Color theme switching
- [ ] Floating emoji background
- [ ] Micro-interactions (inputs, buttons)
- [ ] Typewriter effect (side panel)

### Phase 5: Polish (1 দিন)

- [ ] Toast notifications (funny messages)
- [ ] Responsive design
- [ ] Loading states
- [ ] Error boundaries
- [ ] Testing

**মোট আনুমানিক সময়: 6-9 দিন (একজন mid-level developer এর জন্য)**

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## 💡 Extra Features (Optional / Future)

| Feature | Description |
|---|---|
| **Google OAuth** | "Login with Google (lazy mode 😴)" |
| **Email Verification** | Funny verification email |
| **Remember Me** | "Remember me (pinky promise 🤙)" |
| **Dark/Light Toggle** | Theme switcher |
| **User Dashboard** | After login landing page |
| **Profile Update** | Edit name, avatar |
| **Forgot Password** | Reset via email |

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### 📞 Questions?

ডেভলপার যদি কোনো কিছু unclear মনে হয়, এই README এর প্রতিটি section এ বিস্তারিত আছে। সমস্ত component, API, database schema, এবং animation specification এখানে documented।

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*Made with 🎭 and a questionable sense of humor*

> **"Security is important. But looking good while being secure? That's the real flex."**